

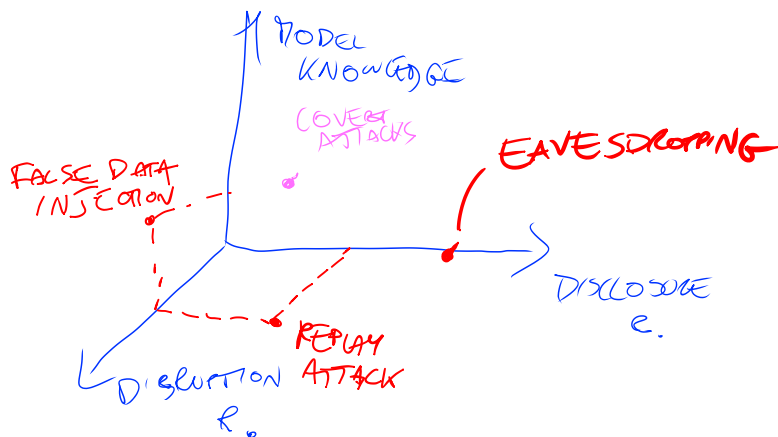
We will study a popular model of the attacker, focusing on attacker resources. We'll divide these resources into three types:

1. Model knowledge: the knowledge of the attacker about the control system (so everything). In particular, the mathematical model of a plant, model, detector.
2. Disclosure resources: the set of measurements and commands an attacker can read/send. Unlike the model knowledge, which is a static knowledge, disclosure resources represent real time information.
3. Disruption resources: the set of measured outputs and commands that the attacker can alter/corrupt/modify in real time. *Example: injecting false data.*

The third one is the one we use for the practical attack.

With the second category, we can estimate the model of the system.

We will study several attacks, looking at a 3d attack space.



- Eavesdropping is just about reading data from communication channels.
- False data injection attack (FDIA): requires model knowledge and disruption resources.
- Replay attack
- Cover attacks: example of an attack requiring all three resource types.

Attacks that are simple to do are close to the origin, while the others are far away.

Course program and definitions:

We will first start with studying the Typical Cyber Physical System - Power System.

Next lesson will be devoted to Risk Management.

Then, we'll see modelling and analysis of cyber physical systems, specifically how to model the plant, the controller and the detector, looking at state space

modelling approach, or Laplace or the z-transformation. The analysis: given a model of a system (the state space model), compute the evolution of variables of interest, given the value of some inputs. For example, if I give a certain a , what is the impact on $x(t)$, or $r(t)$ or $y(t)$ $*_1$. The second step will be to work more in detail with the cyber physical system, so model the networked CPS, introducing and modelling the attacker as well. This is made through the use of an attack logic, an algorithm, that decides in real time what to do, where (which variables), when and how (how to corrupt these variables).

Clever attacks are based on: the careful study of the behaviour of the target system.

$*_1$: the analysis part covers intrinsic properties of the control system, specifically stability and observability. The first one is related more with the impact of the attack, while the second part is related with the goal of remaining undetected. For the modelling of the entire CPS interconnected with the attacker we're going to see two approaches:

- ??

Then, we'll study the fundamental limitations of monitors. We will see when and why undetectable attacks exist.

We use the modelling framework to do model standard types of attacks, such as:

- False data injection attacks, Bias Injection Attacks;
- Denial of Service attacks;
- Replay attacks
- ...

Then, we go on with the study of main cyber physical attacks known from the literature. We start with the False Data Injection Attack against state estimation in power systems.

Imagine we have a power grid as physical system. Operators, from this grid, take power measurements, and those measurements reach a control center. Inside this control center, there will be a server performing state estimation (understanding the current state of the grid, specifically the voltage level across the grid). The output of the state estimation is an estimate to the voltage, $\hat{x}$, such that $\hat{x} \approx x$ (the actual state variables).

We'll have several control algorithms, which will all take the state estimation as input. They'll then take decisions, and send these decisions to the grid.

The false data injection attack was based on injecting data on z , the power

measurements, with a , such that the state estimation receives z_a .

The control center will have, however, a filter for bad data, with bad data detectors.

The goal of the attacker was to determine how, when, where inject data such that the bad data detector will not detect anything strange. To skip detection, the design a must be based on the model of the grid. If the attack is successful, the estimate $\hat{x}$ will be very different from the state x , so $\hat{x} \not\approx x$. This will imply that commands given by control algorithms will not be the good ones. This will lead to disruption of the physical system.

Then, we switch to complex coordinate multilevel attacks.

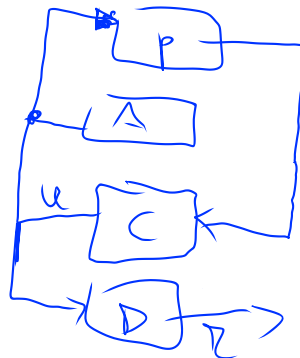
This is an optimisation problem, such that: $\max |f(i)|$, so maximizing the frequency level of a target node i , and as constraints we have ...

After this, we focus on two more physical attacks:

- LOAL altering attacks, the attacker takes control of physical devices, trying to cause problems to the grid.
- Switching attacks, similar, but instead of modulating the power consumption, the attacker connects and disconnects a load, like large-power industries. By doing this in a proper way, we can cause problems to the grid.

The goal of both attacks is to de-stabilise frequency of the grid.

After these attacks, we switch to more sophisticated attacks, which exploit model knowledge and advanced system theory concepts. Specifically, they're zero-dynamics attacks and covert attacks. What is a zero-dynamics attack? We have, as usually, a plant, the output y , the control, the detector, the commands and the residual (the usual model). In the middle we have an attacker, with the attacker only corrupting the commands sent to the plant.



Zero-dynamics attacks answer to this question: find $a(t)$ that have large impact on x , and zero impact on y . This implies there is no impact on the residual r . Thus, the attack is undetectable.

The other type of attack is covert, specifically covert misappropriation of the system. So, an attacker takes control of the Control system. Remember that the whole purpose of the control system is to bring the output close to a desired reference. In general, we have two different control problems:

- Regulation: keep y at desired constant reference.
- Tracking: reference $y^{ref}(t)$ is time varying.

A covert attacks wants to change the state to the system, to a new reference he decides. The attacker will falsify the commands. This will lead to the output deviating from the nominal output, going from $y \rightarrow y_a$. The attacker will also deal with output y , which in the meanwhile has deviated from the reference values, and before y reaches the control block, it will have to be brought back to values close to the reference.

Then, either we go on with Encrypted Control, or control allocation.

We will do simulations in Matlab, Julia, Python.

CPS Power System

The power system is divided into 4 segments:

- Power generation, generation of electricity takes place in big facilities;
- Transmission network, used to connect the power generated by facilities to a distribution network;
- Distribution network, the goal is to take power from big transmission network, and distribute it among small cities, country side, and so on. It works on lower voltages than the two before.
- Consumers.

Electricity follows a sinusoidal function, with frequency oscillating between 50 to 60 Hz.

In a power plant, we have one part dedicated to generate electricity, and we also have a substation for generation, connected to the transmission network. The generation operates typically at 2 kV, but in the substation through transformers the value is raised up to 500 kV.

By looking inside the power plant, we can see several generation units. In the photo, it could be a steam turbine, having the same role as a diesel engine. The engine produces a spin, that is passed to an alternator, an electrical generator.

Electrical power will be forwarded out of the alternator.

What is inside the generation units? Turbines.

On units we may have local controllers and local protectors, effectively computers. The local protection measures the spinning rotation, and if something is wrong disconnects the machine. The local control decides the opening of the valve, in a hydro generation unit.

One important thing is that we have to keep a balance between the power we generate, and the power we consume. The controller does exactly this job. We cannot control the power consumed (we have no power over electricity consumed by customers), but we can try to control the amount of power generated.

Let's go back to the hydro generator. We have to imagine that inside there is a mass rotating, that by spinning generates power. This mass is moved, by one side, by the water entering inside (mechanical power p^M). We also have another amount of power p^e , dependant on what people are asking, which applies a resistance to the rotation.

The rotation is constant when the accelerating power and breaking power are the same.

What happens if someone switches off a load? Now, the machine will accelerate, has the power resisting the acceleration has been reduced. The controller will have to close a bit the valve, reducing the mechanical power of water.

The rotations of the machine could be modelled by the following equations:

$$\begin{aligned}\dot{\delta}_i &= \omega_i \\ M\dot{\omega}_i &= P_i^M - D_i^G\omega_i - P_i^G\end{aligned}$$

This is an example of a state space model. δ is the angle of rotation of the rotor, while ω is the angular velocity.

In state space model, we always have on the left the derivative. $\dot{\delta}$ is the derivate of δ with time, so $\dot{\delta} = \frac{d\delta(t)}{dt}$, the derivative tells us the velocity of how a variable is changing.

Some equations are the statements of definition. $\dot{\delta} = \omega$ tells us the angular velocity is the derivative of the angle rotation.

Whenever we model a physical system, we ask ourselves which are the physical laws which models a system. The second equation is the Newton law for (?).

Remember that Netwon Law states that $F = M \cdot Q$. Acceleration is the derivative of the velocity, so $a = \dot{v} = \frac{dv}{dt}$. So $M\dot{\omega}_i$ is known as the power balance. So M is the

mass, multiplied to the angular velocity.

p^M is the positive mechanical power, p^G is the generated power (the one we put inside the grid, which depends on the consumption), it opposes the movement, acting as a break power. The remaining element $D^G\omega$ is the friction. So we have a balance, having one force trying to accelerate the machine, and the others try to break. When they're in balance, the derivative is close to zero.

The state model thus consists of differential equations. They tell us how variables change in time. We can think the derivate as $\dot{\delta} = \frac{\Delta\delta}{\Delta t}$.

The swing equations models generators.

Now, the other object is the network which interconnects everything.

In all cyber physical systems we'll have local agents, and coupling network.

The grid model is completely different. In the slide, we have a simplified representation of USA grid. We have 14 busses, which are nodes of the network. They're important parts of the network, specifically when there is a generator. Arrows are loads, which could be a city or an entire state. Then, we have cables, which are electrical lines, with arrows on top of them.

We've seen how to model generators, with swing equations. Now, how do we model cables? We have power flow equations. *We do not need to remember them, just understand them.*

They are relations equations, telling us how the power flow from a bus i to a bus j . P is the active power, Q is the reactive power. v is the voltage level, and the equation depends on the voltage level of bus i and j . Remember that voltages are oscillating, and we can think them as the amplitudes. We also have extra parameters of the cable G, B . θ_{ij} is the difference between the phase of j and the phase of i . It's the angular displacement between the two signals, so it's basically a shift. So the P equation makes possible to compute the power flowing depending on voltages, the shift, and other parameters, and it's the line powerflow equation.